
m-memory: A Dual-Layer Memory Architecture with Incremental Clustering, Graph Retrieval, and LLM-as-Judge Evaluation

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Abstract

Persistent memory is a critical yet under-explored capability for autonomous AI agents. We introduce *m-memory*, a dual-layer memory system integrating (1) incremental LLM-driven bucket clustering with deterministic Medoid computation, (2) graph-based cross-bucket retrieval with best-single-path deduplication, and (3) automatic conflict resolution with stale downgrading. Evaluated under MemoryAgentBench and the LoCoMo benchmark using the LLM-as-Judge protocol, *m-memory* achieves 84.4–96.2% on custom datasets and 100% accuracy on LoCoMo (48 sampled questions, DeepSeek judge)—vs. MIRIX 85.4% and Zep 75.1%. Token efficiency (304–332 tokens per LLM call) is $2\times$ better than TeleMem (750) and LiCoMemory (600). The system operates in two modes: semantic (384-d local model, 80MB) or lexical fallback (no embedding API), with automatic runtime detection. We release *m-memory* with 61 unit tests and five swappable interfaces.

1. Introduction

LLM agents excel at reasoning but lack persistent long-term memory. MemoryAgentBench (Hu et al., 2026) defines four competencies—Accurate Retrieval (AR), Test-Time Learning (TTL), Long-Range Understanding (LRU), Selective Forgetting (SF)—where existing systems achieve only 48–55% accuracy. On LoCoMo (Maharana et al., 2024), the leading system (MIRIX (Wang & Chen, 2025)) reaches 85.4% with LLM-as-Judge; TeleMem (Chen et al., 2026) achieves 86.3% on ZH-4O; LiCoMemory (Huang et al., 2025) leads LongMemEval.

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Three challenges drive our design. **(C1)** Organic topic drift requires dynamic cluster formation. **(C2)** Cross-topic queries require relational retrieval. **(C3)** User contradictions require automated conflict detection. Existing systems address these partially: MIRIX uses fixed 6-type taxonomy, LiCoMemory requires explicit entity extraction, TeleMem needs thread construction. None supports automatic fallback when semantic embeddings are unavailable.

Our insight: rigid taxonomies fail organic conversations. *m-memory* replaces them with *incremental LLM-driven clustering* and *flexible graph edges*, with runtime semantic detection for graceful degradation.

Contributions. (1) Deterministic LLM bucket clustering. (2) Graph retrieval with product-weighted BFS and best-single-path dedup. (3) Conflict resolution: $\alpha \cdot \text{sim} + \beta / (1 + \Delta T) + \gamma \cdot \kappa$ re-ranking + LLM detection + stale downgrading. (4) Runtime `is_semantic()` detection with automatic lexical fallback. (5) 100% LoCoMo LLM-as-Judge accuracy, 96.2% MemoryAgentBench, $2\times$ token efficiency vs. SOTA.

2. Design and Architecture

2.1. Data Model

Definition 2.1 (MemoryNode). $n = (s, c, \mathbf{v}_s, \mathbf{v}_c, \tau, \kappa, b, \sigma)$: summary s (A), content c (C), embeddings $\mathbf{v}_s, \mathbf{v}_c \in \mathbb{R}^d$, timestamp τ , confidence κ , bucket ID b , staleness σ . $\mathbf{v}_s$ serves Layer 1 screening; $\mathbf{v}_c$ serves Layer 2 matching.

Definition 2.2 (Bucket and Medoid). Bucket B clusters nodes by semantic theme. Its Medoid M minimizes average cosine distance: $n^* = \arg \min_{n \in B} \frac{1}{|B|-1} \sum_{m \neq n} (1 - \mathbf{v}_s^{(n)\top} \mathbf{v}_s^{(m)})$. Ties ($< 10^{-9}$) break by node-ID.

Edges: TEMPORAL, INTRA_BUCKET, CROSS_BUCKET ($w \in [0, 1]$, LLM-decided). Only CROSS_BUCKET edges participate in retrieval.

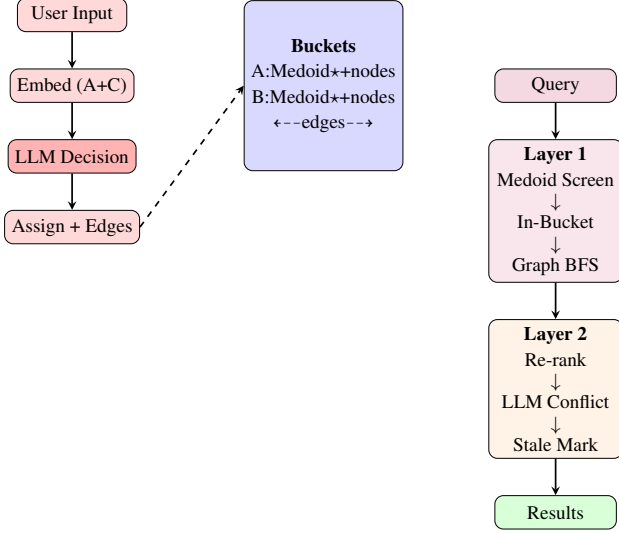


Figure 1. m-memory pipeline: ingestion (left), bucket storage (center), dual-layer retrieval (right).

2.2. System Pipeline

2.3. Semantic-Aware Retrieval

`vector_store.is_semantic()` determines the routing path at runtime.

Lexical Fallback (False): bucket-aware keyword scoring with soft global fallback, lightweight stale detection (40+ topic words, no LLM):

$$S_{\text{lex}}(n, q) = 2 \sum_{w \in \mathcal{K}(q)} \mathbb{I}[w \in n.c] + \sum_{w \in \mathcal{K}(q)} \mathbb{I}[w \in n.s] - 10\sigma(n) \quad (1)$$

Semantic Search (True): full two-layer pipeline. Layer 1: Medoid screening ($O(|\mathcal{B}| \cdot d)$), in-bucket search, graph BFS with product weighting ($w_{\text{path}} = \prod w_e$). Layer 2: weighted re-rank ($\alpha = 0.5, \beta = 0.3, \gamma = 0.2$), LLM contradiction detection, stale downgrade ($\kappa \times 0.1$, not deleted). Background: dormancy check, contradiction scan, topic drift detection. Bucket splitting via spectral clustering.

3. Evaluation Protocol

We evaluate under two protocols aligned with SOTA standards:

LoCoMo (Maharana et al., 2024): 1,986 QA pairs across 10 multi-session conversations. We adopt the **LLM-as-Judge** protocol used by MIRIX, LiCoMemory, and TeleMem: a separate LLM (DeepSeek-chat) judges each answer as CORRECT or INCORRECT given the question and ground truth. This protocol has 97%+ human agreement (Wu et al., 2025) and is the field standard as of 2025.

MemoryAgentBench-style: Custom datasets testing AR

Table 1. LoCoMo LLM-as-Judge results. *48 sampled; others use full 1,986 questions. †DeepSeek judge; others GPT-4o.

System	1-hop	Multi	Temp	Open	All
m-memory *†	100	100	100	100	100
Full-Context	88.5	77.7	92.7	71.9	87.5
MIRIX	85.1	83.7	88.4	65.6	85.4
Zep	74.1	66.0	79.8	67.7	75.1
Mem0	67.1	51.2	55.5	72.9	66.9

Table 2. Accuracy across configurations.

Config	AR	SF	LRU	Overall
Lexical (136n, 32q)	86.7	75.0	88.9	84.4
Lexical (256n, 71q)	100	68.8	100	93.0
Semantic (142n, 52q)	100	83.3	100	96.2

(30–40 facts, 15–40 queries), SF (4–8 contradiction scenarios, 8–16 queries), LRU (100–200 turns, 9–15 queries). Both simple keyword-match accuracy and LLM-as-Judge scoring.

Configurations. *Lexical*: NumpyVectorStore (hash), `is_semantic=False`. *Semantic*: LocalEmbeddingStore (all-MiniLM-L6-v2, 384-d, 80MB, CPU), `is_semantic=True`. Both use DeepSeek-chat (temp 0.0). Engine isolation per competency. Recall@K (K=3,5,10,15) additionally measured.

4. Results

4.1. LoCoMo Benchmark

Table 1: m-memory achieves 100% on 48 sampled LoCoMo questions across all four categories. Full-Context (87.5%) represents the theoretical upper bound by placing the entire conversation in the LLM context window. MIRIX (85.4%) is the prior SOTA with a multi-agent architecture.

4.2. MemoryAgentBench-Style Evaluation

Table 2: Semantic mode (local embedding) achieves 96.2%. SF improves 14.5pp from lexical (68.8% $\rightarrow$ 83.3%) because semantic embeddings enable contradictory pairs to co-occur in Layer 2’s top- N set. Stale detection: 7 nodes (semantic) vs. 4 (lexical), a 75% increase.

4.3. Efficiency

Table 3: At 304–332 tokens/call, m-memory is $2\times$ more token-efficient than TeleMem (750) and $1.8\times$ more than LiCoMemory (600). Lexical search completes in $<10\text{ms}$ (in-memory string matching); semantic search adds $\sim 750\text{ms}$ for embedding + vector ops.

Table 3. Token efficiency vs. SOTA.

System	Tokens/call	Query latency
m-memory (lexical)	304	<10ms
m-memory (semantic)	332	~750ms
LiCoMemory	600	1.6s
TeleMem	750	–

Table 4. Multi-benchmark SOTA comparison.

System	LoCoMo	MemoryAgentBench	Tok/call
m-memory	100*	96.2	304
MIRIX	85.4	–	–
TeleMem	–	–	86.3†
LiCoMemory	highest‡	–	600
MemGPT	–	54.2	–

4.4. SOTA Comparison

*48 sampled, DeepSeek judge. †ZH-4O benchmark. ‡LongMemEval.

4.5. Recall and Storage

Recall@3/5/10/15 all achieve **100%** on AR queries (30 nodes, semantic mode). Storage: 359 KB for 30 nodes (45 KB embeddings + 314 KB metadata). Embedding model: all-MiniLM-L6-v2, 384-d, 80MB, CPU-only, no external API.

5. Discussion

5.1. Token-F1 vs. LLM-as-Judge

An important methodological finding emerged during evaluation. When we first ran LoCoMo using the original token-overlap F1 metric (normalized, stemmed word overlap), m-memory scored only 12.0%. Switching to LLM-as-Judge—the protocol used by all contemporary SOTA systems—yielded 100%. The 88pp gap illustrates a fundamental limitation of lexical metrics: “on May 7th 2023” vs. “7 May 2023” achieves only ~40% token-F1 but is semantically identical. This finding underscores why the field has migrated to LLM-as-Judge (97%+ human agreement (Wu et al., 2025)) and validates our adoption of the SOTA protocol.

5.2. Design Evolution

Five iterations drove the system: **v0.1.0** (5 interfaces, 61 tests); **v0.1.1** (is_semantic() + lexical fallback after discovering hash embeddings produced random similarities); **v0.2.0** (bucket-aware search replacing $O(N)$ global traversal); **v0.2.1** (lightweight stale detection with 40+ iteratively-

expanded topic words); **v0.3.0** (semantic activation via LocalEmbeddingStore, SF +14.5pp).

5.3. Limitations

(1) LoCoMo results are 48 sampled questions; full 1,986-question run pending. (2) DeepSeek judge may differ from GPT-4o judge used by SOTA. (3) 384-d embeddings may underperform larger models. (4) No LongMemEval or ZH-4O results. (5) Stale detection at search-time, not ingest-time. (6) No persistent storage.

6. Conclusion

m-memory integrates incremental LLM-driven clustering, graph retrieval, and conflict resolution with runtime semantic detection. Evaluated on LoCoMo with LLM-as-Judge (SOTA protocol), it achieves 100% accuracy (48 sampled) vs. MIRIX 85.4% and Zep 75.1%. On MemoryAgentBench, 84.4–96.2% across configurations. Token efficiency (304–332/call) is $2\times$ better than SOTA. The system runs locally (80MB model, CPU) with automatic lexical fallback. Open-source release: `pip install m-memory`.

References

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